

Solution Requirements

Date	25 June 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely log in using email and password before accessing the application.	High
2	Authorization Levels	Students can record audio, view scores, and download reports. Administrators can manage users and view evaluation records.	High
3	External Interfaces	Integrates Speech-to-Text API, AI evaluation model, MySQL database, and Streamlit web interface.	High
4	Transaction Processing	The system records voice input, converts it to text, evaluates concept understanding, stores results, and	High

		generates feedback.	
5	Reporting	Generates evaluation score, strengths, weaknesses, transcript, visual charts, and downloadable PDF reports.	High
6	Business Rules	Audio must be clear and within the allowed duration. AI evaluates based on concept accuracy, clarity, fluency, and completeness.	Medium
7	Compliance to Laws or Regulations	User data is stored securely and used only for educational purposes while maintaining data privacy.	Medium
8	Other	Supports repeated evaluations so students can track learning progress over time.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process speech quickly and generate results without noticeable delay.	Feedback generated within 5–10 seconds
2	Scalability	The system should support multiple students using the application simultaneously.	Supports 100+ concurrent users
3	Security & Data Privacy	User data, transcripts, and reports should be securely stored with authenticated access.	Password-protected login and encrypted database communication
4	Reliability & Availability	The application should remain available and provide consistent evaluation results.	99% system availability

5	Usability & Accessibility	The interface should be simple, responsive, and easy for students and teachers to use.	Easy navigation with responsive UI across devices
6	Other	The system should be maintainable and portable for future enhancements and deployment.	Modular code with support for future AI model upgrades